Ellan Vannin Wealth Management

Sovereign Infrastructure Investment Memorandum

The Dee Estuary Renewable Energy Programme

A Strategic Tidal & Hydro Energy Investment for the Liverpool–North Wales Growth Corridor

Prepared for:
Ellan Vannin Wealth Management (EVWM)

Executive Summary

Ellan Vannin Wealth Management proposes the creation of the **Dee Estuary Renewable Energy Programme (DEREP)**, a sovereign-style infrastructure investment designed to transform the River Dee estuary into one of Britain's principal renewable energy and coastal resilience assets.

Unlike the larger Morecambe Bay proposal, the Dee programme would focus on a combination of:

- Tidal barrage generation
- Tidal stream turbines
- River hydro generation upstream
- Flood defence
- Smart grid infrastructure
- Green hydrogen production
- Industrial power supply for North Wales and the Liverpool City Region

Historic engineering studies have proposed tidal barrages across the Dee for over a century, and UK offshore energy assessments continue to identify the Dee as one of the UK's significant tidal-range resources, with an indicative potential of around **800 MW**.

Strategic Vision

The Dee Estuary already sits between two major economic regions:

Liverpool City Region

- Liverpool
- Wirral

and

North Wales

- Flintshire
- Deeside
- Chester
- Wrexham
- North Wales Coast

The objective is to create a renewable energy corridor supporting both economies while improving flood resilience and industrial competitiveness.

Project Components

Phase One

Dee Estuary Tidal Barrage

Indicative location:

Between the Wirral Peninsula and the Welsh shoreline, positioned to minimise impacts on navigation while capturing tidal energy.

Potential capacity:

Item	Estimate
Installed Capacity	**600–800 MW**
Annual Generation	**1.8–2.5 TWh**
Asset Life	120+ years

UK resource assessments continue to identify the Dee as one of the country's larger tidal-range opportunities.

Phase Two

Tidal Stream Turbine Arrays

Instead of relying solely on a barrage, install underwater tidal turbines within faster-flowing channels.

Advantages:

- Lower ecological impact
- Modular expansion
- Lower initial capital cost

- Continuous technological upgrades

Phase Three

River Dee Hydro Programme

The River Dee above Chester contains numerous historic weirs and controlled water structures.

Rather than constructing large dams, EVWM proposes:

- Low-head hydro turbines
- Archimedes screw generators
- Fish-friendly hydro systems
- Water authority partnerships

Potential locations include existing weirs and navigation structures where environmental approvals could be more achievable.

Indicative capacity:

15–30 MW

Annual generation:

60–120 GWh

Phase Four

Green Hydrogen Production

Use surplus tidal electricity to power electrolysers located near:

- Deeside Industrial Park
- Mostyn Port
- Liverpool docks

Applications:

- Heavy industry
- Shipping
- Bus fleets
- Industrial heat

Electricity Generation

Technology	Annual Output
Tidal barrage	1.8–2.5 TWh

Technology	Annual Output
Tidal stream	0.3–0.5 TWh
River hydro	0.06–0.12 TWh

Total Annual Generation

2.2–3.1 TWh

Equivalent to approximately **600,000–850,000 average UK homes**, depending on household consumption assumptions.

Economic Impact

Energy Security

Supports:

- Liverpool
- Deeside Industrial Zone
- Cheshire
- North Wales

The Dee Industrial Corridor contains major manufacturers in aerospace, automotive, food processing and advanced manufacturing that could benefit from long-term low-carbon electricity supplies.

Industrial Benefits

Direct beneficiaries include:

- Deeside Industrial Park
- Airbus Broughton
- Port of Mostyn
- Liverpool logistics sector
- Hydrogen production
- Data centres

GDP Impact

Illustrative regional economy served:

Approximately:

£70–80 billion GVA

Illustrative long-term effects:

Benefit	Annual Estimate
Industrial productivity	£350–600m
Energy cost savings	£150–250m
Construction supply chain	£200m
Hydrogen economy	£150–300m

Total Potential Economic Benefit

£850 million–1.35 billion annually

These are strategic appraisal estimates rather than forecasts.

Capital Investment

Component	Estimated Cost
Tidal barrage	£3.5–5.0bn
Tidal stream arrays	£600m
River hydro installations	£200m
Hydrogen facilities	£700m
Grid reinforcement	£800m

Total Programme

£6–7.5 billion

Revenue Streams

Illustrative annual revenues:

Source	Annual
Electricity sales	£180–300m
Grid balancing	£20–40m
Hydrogen production	£80–150m
Renewable certificates	£20–40m

Total Annual Revenue

£300–530 million

Actual revenues would depend on wholesale electricity prices, hydrogen demand and the final project configuration.

Environmental Strategy

Unlike historic concepts centred on a single large barrage, EVWM recommends a **hybrid approach**:

- Smaller tidal impoundments where appropriate
- Tidal stream turbines in high-flow channels
- Fish passage systems
- Wetland creation
- Saltmarsh restoration
- Flood defence integration

Recent expert reviews of UK tidal energy have increasingly favoured flexible combinations of tidal technologies rather than relying exclusively on very large barrages because they can better balance energy generation with environmental considerations.

Investment Recommendation

Ellan Vannin Wealth Management should establish a **Dee Estuary Renewable Infrastructure Fund** as part of its wider Irish Sea investment strategy.

The project would complement the proposed:

- North Wales–Liverpool Growth Corridor
- Liverpool–Douglas Irish Sea Express
- Fylde–Furness Tidal Gateway
- Irish Sea Northern Spine Railway

Together, these projects would form an integrated Irish Sea infrastructure portfolio centred on clean energy, transport and long-term regional productivity.

The Dee programme offers a comparatively lower-cost, modular renewable energy investment than the larger Morecambe Bay concept, while still delivering significant economic benefits, supporting industrial decarbonisation and strengthening the Liverpool–North Wales growth corridor.